

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: Software: System & Application

Introduction

- Software is the set of programs that instructs the hardware. It is divided into system software (which runs the computer) and application software (which does the user's work).

System software

- **Operating system** — manages resources and provides the interface (Windows, macOS, Linux, Android, iOS).
- **Utility software** — maintenance tasks: antivirus, disk cleanup, compression, backup, defragmentation.
- **Device drivers** — allow the OS to communicate with hardware (printer drivers, graphics drivers).
- **Language translators** — compilers, interpreters and assemblers that convert program code into machine language.

Application software

- **Word processing** — Word, Google Docs; **spreadsheets** — Excel; **presentations** — PowerPoint; **databases** — Access, Oracle; **graphics** — Photoshop, CorelDRAW; **browsers** — Chrome, Firefox; **media players and games**. Application software may be general-purpose, special-purpose or tailor-made/custom software.

Software categories by licensing

- Proprietary (purchased, closed source — Windows, Microsoft Office); open-source (source code available and free — Linux, LibreOffice); freeware (free, but closed — Skype); shareware (free trial, pay to continue).

CLASS EVALUATION

1. Define software.
2. Differentiate between system and application software with examples.
3. What is a device driver?

4. Open-source software is free of charge but not necessarily free of restrictions — explain.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: Operating Systems (Basics)

Introduction

- The operating system (OS) is the master program that manages the computer's hardware and software resources and provides the user interface.

Functions of an operating system

- **Memory management** — allocates RAM to programs; **processor (process) management** — schedules programs to use the CPU; **file management** — creates, names, organises and protects files into folders; **device management** — controls input/output devices through drivers; **security** — user accounts, passwords, permissions; **user interface** — command-line (CLI) or graphical (GUI); **networking and multitasking**.

Types of operating systems

- **Single-user, single-task** — one user, one program at a time (early MS-DOS).
- **Single-user, multi-tasking** — one user running many programs (Windows, macOS).
- **Multi-user** — many users share one system (UNIX, Linux servers).
- **Real-time** — immediate response (aircraft control, ATMs); **network** and **distributed** OSes manage connected systems; **mobile OSes** — Android, iOS.

Interfaces

- CLI: the user types commands (MS-DOS, Linux terminal) — powerful but requires memorising commands. GUI: windows, icons, menus and a pointer (WIMP) — user-friendly, as in Windows. Modern systems offer both.

CLASS EVALUATION

1. Define an operating system.
2. State four functions of an OS.
3. Differentiate between CLI and GUI.
4. Give one example of a real-time operating system.

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WEEK: NINE

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: Storage & Memory

Introduction

- Memory and storage hold the data and programs the computer uses. Memory is the workspace of the CPU (primary storage); storage is the permanent holding area (secondary storage).

Primary memory

- **RAM** — random access memory: read/write, volatile, temporary; more RAM means more programs can run smoothly.
- **ROM** — read-only memory: permanent instructions (the boot program/BIOS); non-volatile.
- **Cache** — very fast memory close to the CPU holding frequently used data; **registers** — the fastest, inside the CPU.
- Units: bit → byte (8 bits) → KB (1024 B) → MB → GB → TB.

Secondary storage

- **Magnetic** — hard disk drive (HDD): large, cheap, with moving parts.
- **Solid state** — SSD and flash drives: faster, no moving parts, shock-resistant but costlier per gigabyte.
- **Optical** — CD (700 MB), DVD (4.7 GB), Blu-ray (25 GB+).
- **Cloud storage** — remote servers (Google Drive, OneDrive): accessible anywhere, but needs the internet.

Choosing storage

- Consider capacity, speed (access time), portability, durability, cost, and the importance of the data (always keep a backup in a second location).

CLASS EVALUATION

1. Differentiate between RAM and ROM.
2. Arrange in increasing order: GB, KB, TB, MB.
3. State two advantages of an SSD over an HDD.

4. What is meant by volatile memory?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: File Management

Introduction

- File management is the organisation and control of files and folders on a storage device — how files are named, stored, arranged, copied, moved and protected.

Files and folders

- A file is a named unit of data (document, picture, program); files are stored in folders (directories), which can contain sub-folders. Every file has a name, an extension (which shows the type — .docx, .jpg, .mp4, .exe), a size, a type and a date modified.

Operations

- Create, rename, copy, move (cut and paste), delete (to the Recycle Bin/Trash), restore, search and sort (by name, size, date, type); compress files (zip) to save space and make sharing easier; shortcut icons link to files without duplicating them.

Good practices

- Use clear folder structures (Documents → School → Mathematics); name files descriptively; save regularly; work on a copy of important files; empty the Recycle Bin periodically; protect files with passwords where needed; and back up to a second drive or the cloud.

CLASS EVALUATION

1. Define a file and a folder.
2. What does the extension .docx tell you?
3. State three file-management operations.
4. Why is it good practice to back up files?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: The Internet & WWW

Introduction

- The Internet is the global network of interconnected computers using the TCP/IP protocol suite; the World Wide Web (WWW) is the system of interlinked documents (web pages) accessible over the Internet.

How it works

- Data is moved in packets; each computer has a unique IP address; the Domain Name System (DNS) translates human-readable names (www.example.ng) into IP addresses. A web browser (Chrome, Edge, Firefox) requests pages from web servers using HTTP/HTTPS; the pages are written in HTML. Search engines (Google, Bing) index the web and return results for queries.

Services of the Internet

- The Web (websites), e-mail, file transfer (FTP), social media, instant messaging (WhatsApp), video conferencing, e-commerce, e-banking, online learning and the Internet of Things.

The browser and URLs

- A URL (Uniform Resource Locator) identifies a resource: <https://www.example.com/page> — scheme, domain and path. Browser tools: address bar, tabs, bookmarks, history, downloads and privacy settings (cookies may track users).

CLASS EVALUATION

1. Differentiate between the Internet and the WWW.
2. What is the function of DNS?
3. Name three services of the Internet.
4. State the parts of a URL with an example.

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WEEK: TWELVE

CLASS: SS1

SUBJECT: Digital Technology

TOPIC: E-Mail & Electronic Communication

Introduction

- E-mail (electronic mail) is the sending and receiving of messages over the Internet. It is fast, cheap, records a copy of the message, and can carry attachments — but has limits on size and can be abused.

E-mail basics

- An address has the form username@domain (e.g. ada@mail.com): the part before @ is the user name, after it the mail server. Services: webmail (accessed in a browser — Gmail, Yahoo) and desktop/mobile clients (Outlook). Components of a message: To, Cc, Bcc, Subject, the body, and attachments.

Netiquette and safety

- Write clear subject lines; be polite and concise; do not type in capitals (shouting); use Cc appropriately and Bcc for privacy; send attachments only when needed (size limits apply); think before sending; and never open suspicious attachments or click links from unknown senders (phishing).

Other electronic communication tools

- SMS/MMS, instant messaging (WhatsApp, Telegram), social media, voice over IP (Zoom, Teams, Skype), and discussion forums — each with its etiquette.

CLASS EVALUATION

1. What is e-mail?
2. State the parts of an e-mail address.
3. What does Bcc do?
4. Give three rules of e-mail etiquette.